16
THE DOUBLE SLIT

The interference of light from two narrow slits close together, first demonstrated by Young, has already been discussed (Sec. 13.2) as a simple example of the interference of two beams of light. In our discussion of the experiment, the slits were assumed to have widths not much greater than a wavelength of light, so that the central maximum in the diffraction pattern from each slit separately was wide enough to occupy a large angle behind the screen (Figs. 13A and 13B). It is important to understand the modifications of the interference pattern which occur when the width of the individual slits is made greater, until it becomes comparable with the distance between them. This corresponds more nearly to the actual conditions under which the experiment is usually performed. In this chapter we shall discuss the *Fraunhofer diffraction by a double slit,* and some of its applications.

16.1 QUALITATIVE ASPECTS OF THE PATTERN

In Fig. 16A(*b*) and (*c*) photographs are shown of the patterns obtained from two different double slits in which the widths of the individual slits were equal in each pair but the two pairs were different. Figure 16B shows the experimental arrangement for

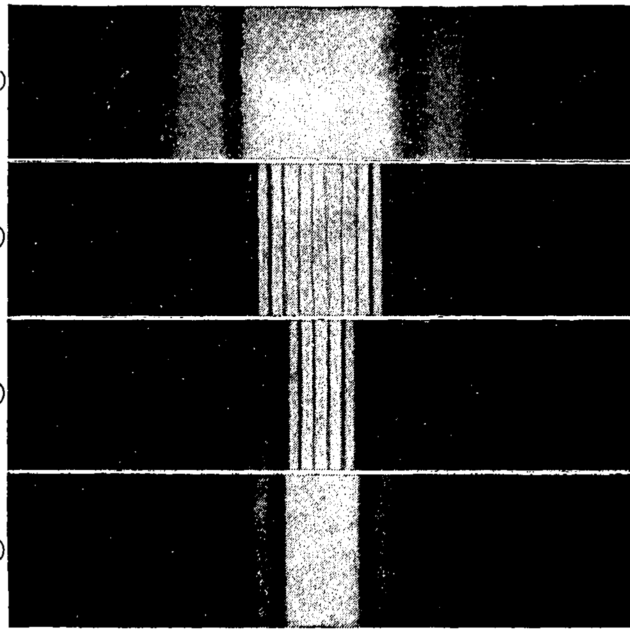

FIGURE 16A
Diffraction patterns from (*a*) a single narrow slit, (*b*) two narrow slits, (*c*) two wider slits, (*d*) one wider slit.

photographing these patterns; the *slit width b* of each slit was greater for Fig. 16A(*c*) than for Fig. 16A(*b*), but the distance between centers $d = b + c$, or the *separation* of the slits, was the same in the two cases. In the central part of Fig. 16A(*b*) are seen a number of interference maxima of approximately uniform intensity, resembling the interference fringes described in Chap. 13 and shown in Fig. 13D. The intensities of these maxima are not actually constant, however, but fall off slowly to zero on either side and then reappear with low intensity two or three times before becoming too faint to observe without difficulty. The same changes are seen to occur much more rapidly in Fig. 16A(*c*), which was taken with the slit widths b somewhat larger.

16.2 DERIVATION OF THE EQUATION FOR THE INTENSITY

Following the same procedure as that used for the single slit in Sec. 15.2, it is merely necessary to change the limits of integration in Eq. (15b) to include the two portions of the wave front transmitted by the double slit.* Thus if we have, as in Fig. 16B, two equal slits of width b, separated by an opaque space of width c, the origin may

* The result of this derivation is obviously a special case of the general formula for N slits, which will be obtained by the method of complex amplitudes in the following chapter.

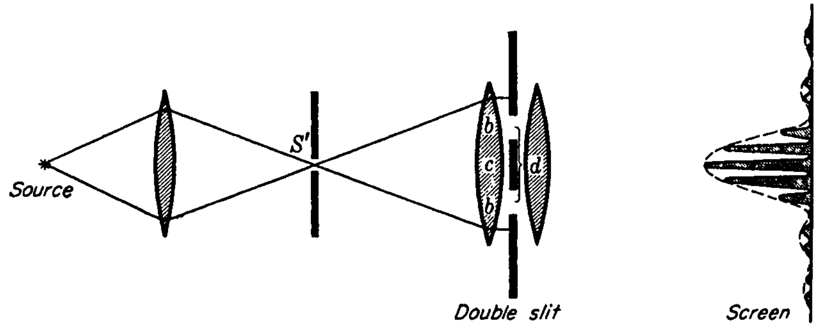

FIGURE 16B
Apparatus for observing Fraunhofer diffraction from a double slit. Drawn for $2b = c$, that is, $d = 3b$.

be chosen at the center of c, and the integration extended from $s = d/2 - b/2$ to $s = d/2 + b/2$. This gives

$$y = \frac{2a}{xk \sin \theta} \{\sin [\tfrac{1}{2}k(d + b) \sin \theta] - \sin [\tfrac{1}{2}k(d - b) \sin \theta]\} [\sin (\omega t - kx)]$$

The quantity in braces is of the form $\sin (A + B) - \sin (A - B)$, and when it is expanded, we obtain

$$y = \frac{2ba}{x} \frac{\sin \beta}{\beta} \cos \gamma \sin (\omega t - kx) \tag{16a}$$

where, as before,

$$\beta = \tfrac{1}{2}kb \sin \theta = \frac{\pi}{\lambda} b \sin \theta$$

and where

$$\gamma = \tfrac{1}{2}k(b + c) \sin \theta = \frac{\pi}{\lambda} d \sin \theta \tag{16b}$$

The intensity is proportional to the square of the amplitude of Eq. (16a), so that, replacing ba/x by A_0 as before, we have

● $$I = 4A_0^2 \frac{\sin^2 \beta}{\beta^2} \cos^2 \gamma \tag{16c}$$

The factor $(\sin^2 \beta)/\beta^2$ in this equation is just that derived for the single slit of width b in the previous chapter [Eq. (15d)]. The second factor $\cos^2 \gamma$ is characteristic of the interference pattern produced by two beams of equal intensity and phase difference δ, as shown in Eq. (13b) of Sec. 13.3. There the resultant intensity was found to be proportional to $\cos^2 (\delta/2)$, so that the expressions correspond if we put $\gamma = \delta/2$. The resultant intensity will be zero when either of the two factors is zero. For the first factor this will occur when $\beta = \pi, 2\pi, 3\pi, \ldots$, and for the second factor when $\gamma = \pi/2, 3\pi/2, 5\pi/2, \ldots$. That the two variables β and γ are not independent will be seen from Fig. 16C. The difference in path from the two edges of a given slit to the screen is, as indicated, $b \sin \theta$. The corresponding phase difference is, by Eq. (15c),

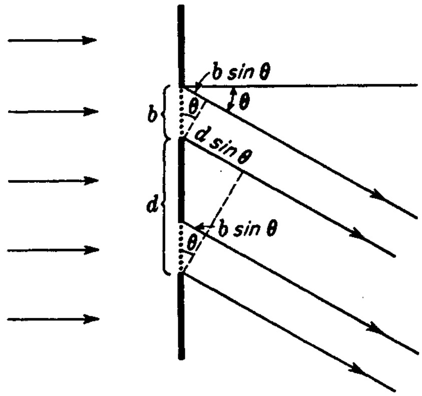

FIGURE 16C
Path differences of parallel rays leaving a double slit.

$(2\pi/\lambda)b$ sin θ, which equals 2β. The path difference from any two corresponding points in the two slits, as illustrated for the two points at the lower edges of the slits, is $d \sin\theta$, and the phase difference is $\delta = (2\pi/\lambda)d \sin\theta = 2\gamma$. Therefore, in terms of the dimensions of the slits,

$$\frac{\delta}{2\beta} = \frac{\gamma}{\beta} = \frac{d}{b} \qquad \text{(16d)}$$

16.3 COMPARISON OF THE SINGLE-SLIT AND DOUBLE-SLIT PATTERNS

It is instructive to compare the double-slit pattern with that given by a single slit of width equal to that of either of the two slits. This amounts to comparing the effect obtained with the two slits in the arrangement shown in Fig. 16B with that obtained when one of the slits is entirely blocked off with an opaque screen. If this is done, the corresponding single-slit diffraction patterns are observed, and they are related to the double-slit patterns as shown in Fig. 16A(*a*) and (*d*). It will be seen that the intensities of the interference fringes in the double-slit pattern correspond to the intensity of the single-slit pattern at any point. If one or other of the two slits is covered, we obtain exactly the same single-slit pattern in the same position, while if both slits are uncovered, the pattern, instead of being a single-slit one with twice the intensity, breaks up into the narrow maxima and minima called *interference fringes.* The intensity at the maximum of these fringes is 4 times the intensity of either single-slit pattern at that point, while it is zero at the minima (see Sec. 13.4).

16.4 DISTINCTION BETWEEN INTERFERENCE AND DIFFRACTION

One is quite justified in explaining the above results by saying that the light from the two slits undergoes interference to produce fringes of the type obtained with two beams but that the intensities of these fringes are limited by the amount of light arriving at the given point on the screen by virtue of the diffraction occurring at each slit. The relative intensities in the resultant pattern as given by Eq. (16c) are just those obtained by multiplying the intensity function for the interference pattern from two infinitely narrow slits of separation d [Eq. (13b)] by the intensity function for diffraction from a single slit of width b [Eq. (15d)]. Thus, the result may be regarded as due to the joint action of interference between the rays coming from corresponding points in the two slits and of diffraction, which determines the amount of light emerging from either slit at a given angle. But diffraction is merely the result of the interference of all the secondary wavelets originating from the different elements of the wave front. Hence it is proper to say that the whole pattern is an interference pattern. It is just as correct to refer to it as a diffraction pattern, since, as we saw from the derivation of the intensity function in Sec. 16.2, it is obtained by direct summing the effects of all of the elements of the exposed part of the wave front. However, if we reserve the term *interference* for those cases in which a modification of amplitude is produced by the superposition of a finite (usually small) number of beams, and *diffraction* for those in which the amplitude is determined by an integration over the infinitesimal elements of the wave front, the double-slit pattern can be said to be due to a combination of interference and diffraction. Interference of the beams from the two slits produces the narrow maxima and minima given by the $\cos^2 \gamma$ factor, and diffraction, represented by $(\sin^2 \beta)/\beta^2$, modulates the intensities of these interference fringes. The student should not be misled by this statement into thinking that diffraction is anything other than a rather complicated case of interference.

16.5 POSITIONS OF THE MAXIMA AND MINIMA. MISSING ORDERS

As shown in Sec. 16.2, the intensity will be zero wherever $\gamma = \pi/2, 3\pi/2, 5\pi/2, \ldots$ and also when $\beta = \pi, 2\pi, 3\pi, \ldots$. The first of these two sets are the minima for the interference pattern, and since by definition $\gamma = (\pi/\lambda)d \sin \theta$, they occur at angles θ such that

• $$d \sin \theta = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \ldots = (m + \tfrac{1}{2})\lambda \qquad \textit{Minima} \qquad (16e)$$

m being any whole number starting with zero. The second series of minima are those for the diffraction pattern, and these, since $\beta = (\pi/\lambda)a \sin \theta$, occur where

• $$b \sin \theta = \lambda, 2\lambda, 3\lambda, \ldots = p\lambda \qquad \textit{Minima} \qquad (16f)$$

the smallest value of p being 1. The exact positions of the *maxima* are not given by any simple relation, but their approximate positions can be found by neglecting the variation of the factor $(\sin^2 \beta)/\beta^2$, a justified assumption only when the slits are very narrow and when the maxima near the center of the pattern are considered [Fig. 16A(*b*)]. The positions of the maxima will then be determined solely by the $\cos^2 \gamma$ factor, which has maxima for $\gamma = 0, \pi, 2\pi, \ldots$, that is, for

$$d \sin \theta = 0, \lambda, 2\lambda, 3\lambda, \ldots = m\lambda \qquad \textit{Maxima} \qquad (16g)$$

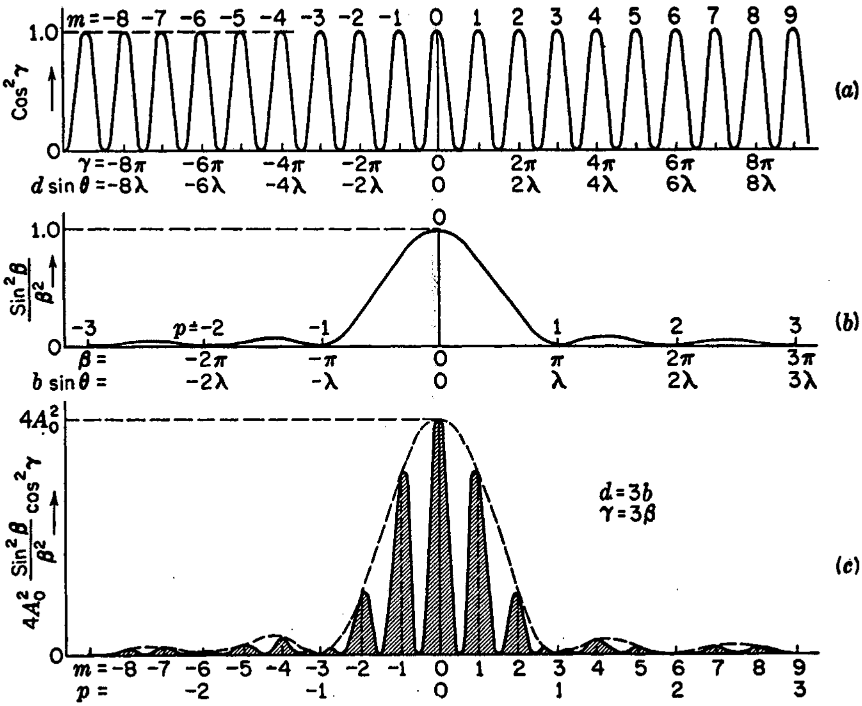

FIGURE 16D
Intensity curves for a double slit where $d = 3b$.

The whole number m represents physically the number of wavelengths in the path difference from corresponding points in the two slits (see Fig. 16C) and represents the *order* of interference.

Figure 16D(a) is a plot of the factor $\cos^2 \gamma$, and here the values of the order, of half the phase difference $\gamma = \delta/2$, and of the path difference are indicated for the various maxima. These are all of equal intensity and equidistant on a scale of $d \sin \theta$, or practically on a scale of θ, since when θ is small, $\sin \theta \approx \theta$ and the maxima occur at angles $\theta = 0, \lambda/d, 2\lambda/d, \ldots$. With a finite slit width b the variation of the factor $(\sin^2 \beta)/\beta^2$ must be taken into account. This factor alone gives just the single-slit pattern discussed in the last chapter, and is plotted in Fig. 16D(b). The complete double-slit pattern as given by Eq. (16c) is the product of these two factors and therefore is obtained by multiplying the ordinates of curve (a) by those of curve (b) and the constant $4A_0{}^2$. This pattern is shown in Fig. 16D(c). The result will depend on the relative scale of the abscissas β and γ, which in the figure are chosen so that for a given abscissa $\gamma = 3\beta$. But the relation between β and γ for a given angle θ is determined, according to Eq. (16d), by the ratio of the slit width to the slit separation. Hence if $d = 3b$, the two curves (a) and (b) are plotted to the same scale of θ. For the particular case of two slits of width b separated by an opaque space of width $c = 2b$, the curve (c), which is the product of (a) and (b), then gives the resultant pattern. The positions of the maxima in this curve are slightly different from those in curve (a) for all except the central maximum ($m = 0$), because when the ordinates near one of the maxima of curve (a) are multiplied by a factor which is decreasing or increasing, the ordinates on one side of the maximum are changed by a different amount from those of the other, and this displaces the resultant maximum slightly in the direction in which the factor is increasing. Hence the positions of the maxima in curve (c) are not exactly those given by Eq. (16g) but in most cases will be very close to them.

Let us now return to the explanation of the differences in the two patterns (b) and (c) of Fig. 16A, taken with the same slit separation d but different slit widths b. Pattern (c) was taken for the case $d = 3b$, and is seen to agree with the description given above. For pattern (b), the slit separation d was the same, giving the same spacing for the interference fringes, but the slit width b was smaller, such that $d = 6b$. In Fig. 13D, $d = 14b$. This greatly increases the scale for the single-slit pattern relative to the interference pattern, so that many interference maxima now fall within the central maximum of the diffraction pattern. Hence the effect of decreasing b, keeping d unchanged, is merely to broaden out the single-slit pattern, which acts as an envelope of the interference pattern as indicated by the dotted curve of Fig. 16D(c).

If the slit-width b is kept constant and the separation of the slits d is varied, the scale of the interference pattern varies, but that of the diffraction pattern remains the same. A series of photographs taken to illustrate this is shown in Fig. 16E. For each pattern three different exposures are shown, to bring out the details of the faint and the strong parts of the pattern. The maxima of the curves are labeled by the order m, and underneath the upper one is a given scale of angular positions θ. A study of these figures shows that certain orders are missing, or at least reduced to two maxima of very low intensity. These *missing orders* occur where the condition for a maximum of the interference, Eq. (16g), and for a minimum of the diffraction, Eq. (16f), are both fulfilled for the same value of θ, that is for

$$d \sin \theta = m\lambda \qquad \text{and} \qquad b \sin \theta = p\lambda$$

so that

$$\frac{d}{b} = \frac{m}{p} \tag{16h}$$

Since m and p are both integers, d/b must be in the ratio of two integers if we are to have missing orders. This ratio determines the orders which are missing, in such a way that when $d/b = 2$, orders 2, 4, 6, ... are missing; when $d/b = 3$, orders 3, 6, 9, ... are missing; etc. When $d/b = 1$, the two slits exactly join, and all orders should be missing. However, the two faint maxima into which each order is split can then be shown to correspond exactly to the subsidiary maxima of a single-slit pattern of width $2b$.

Our physical picture of the cause of missing orders is as follows. Considering, for example, the missing order $m = +3$ in Fig. 16D(c), this point on the screen is just three wavelengths farther from the center of one slit than from the center of the other. Hence we might expect the waves from the two slits to arrive in phase and to produce a maximum. However, this point is at the same time one wavelength farther from the edge of one slit than from the other edge of that slit. Addition of the second-

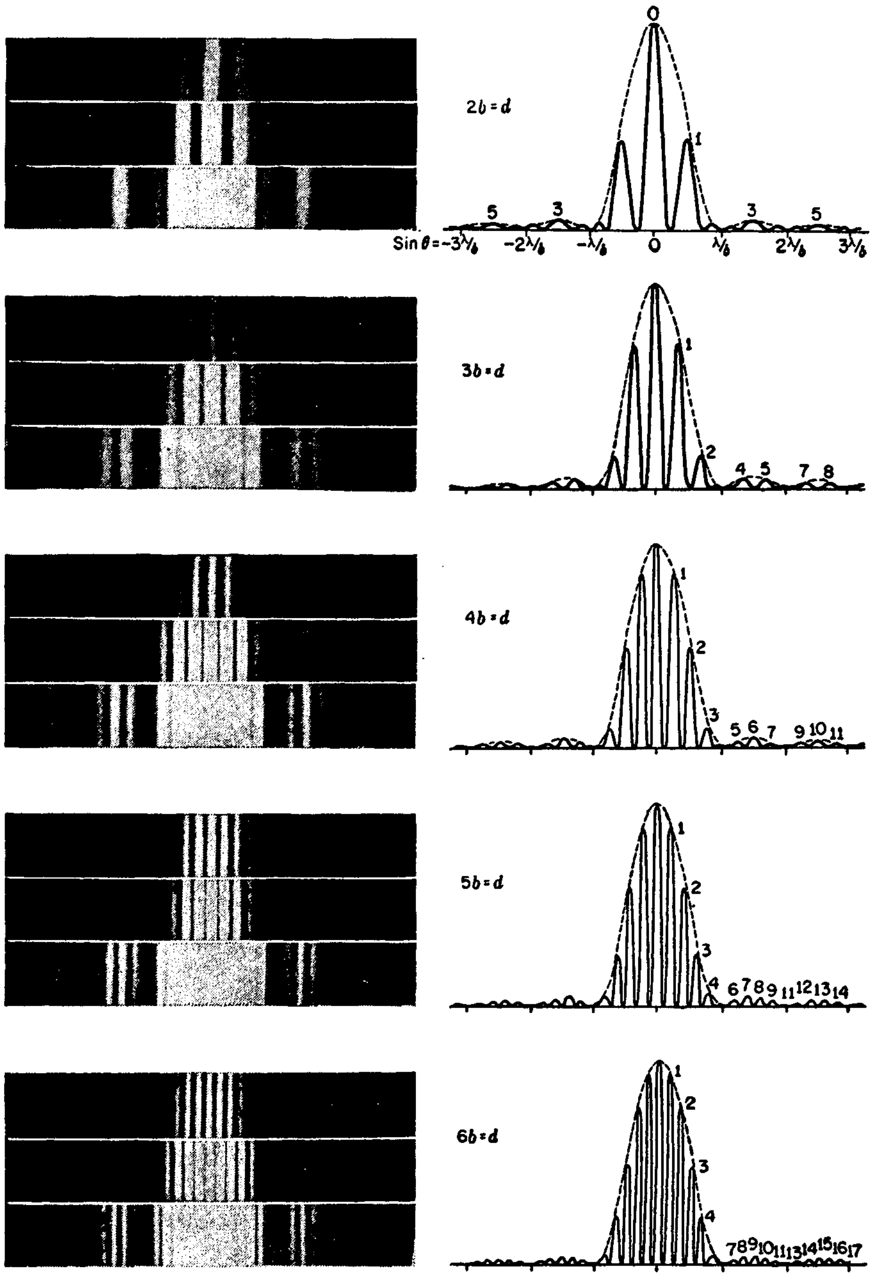

FIGURE 16E
Photographs and intensity curves for double-slit diffraction patterns.

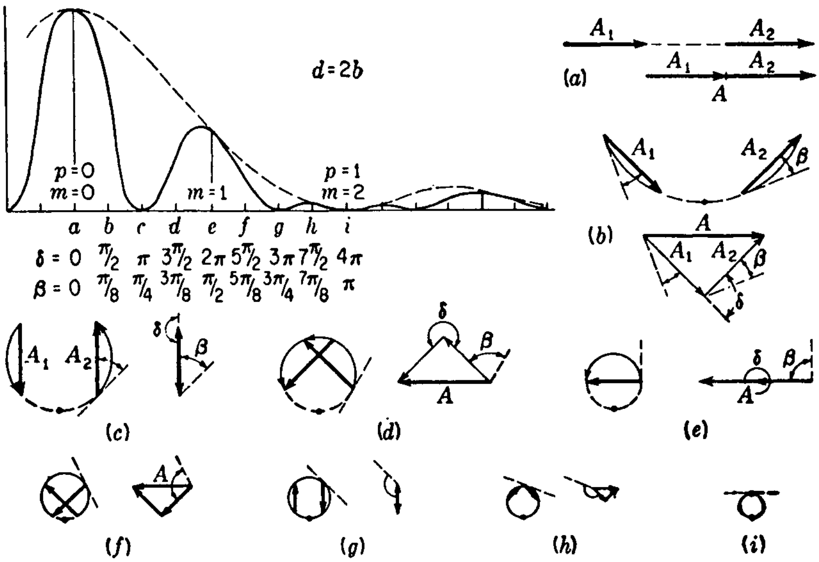

FIGURE 16F
How the intensity curve for a double slit is obtained by the graphical addition of amplitudes.

ary wavelets from one slit gives zero intensity under these conditions. The same holds true for either slit, so that although we may add the contributions from the two slits, both contributions are zero and must therefore give zero resultant.

16.6 VIBRATION CURVE

The same method as that applied in Sec. 15.4 for finding the resultant amplitude graphically in the case of the single slit is applicable to the present problem. For illustration we take a double slit in which the width of each slit equals that of the opaque space between the two, so that $d = 2b$. A photograph of this pattern appears in Fig. 16E at the top. A vector diagram of the amplitude contributions from one slit gives the arc of a circle, as before, the difference between the slopes of the tangents to the arc at the two ends being the phase difference 2β between the contributions from the two edges of the slit. Such an arc must now be drawn for each of the two slits, and the two arcs must be related in such a way that the phases (slopes of the tangents) differ for corresponding points on the two slits by 2γ, or δ. In the present case, since $d = 2b$, we must have $\gamma = 2\beta$ or $\delta = 4\beta$. Thus in Fig. 16F(b) showing the vibration curve for $\beta = \pi/8$, both arcs subtend an angle of $\pi/4$ ($= 2\beta$), the phase difference for the two edges of each slit, and the arcs are separated by $\pi/4$ so that corresponding points on the two arcs differ by $\pi/2(= \delta)$. Now the resultant contributions from the two slits are represented in amplitude and phase by the chords of these two arcs, that is by A_1 and A_2. Diagrams (a) to (i) give the construction for the points similarly labeled on the intensity curve. The intensity, it will be remembered, is found as the square of the resultant amplitude A, which is the vector sum of A_1 and A_2.

In the example chosen, the slits are relatively wide compared with their separation, and as the phase difference increases the curvature of the individual arcs of the vibration curve increases rapidly, so that the vectors A_1 and A_2 decrease rapidly in length. For narrower slits we obtain a greater number of interference fringes within the central diffraction maximum, because the lengths of the arcs are smaller relative to the radius of curvature of the circle. A_1 and A_2 then decrease in length more slowly with increasing β, and the intensities of the maxima do not fall off so rapidly. In the limit where the slit width a approaches zero, A_1 and A_2 remain constant, and the variation of the resultant intensity is merely due to the change in phase angle between them.

16.7 EFFECT OF FINITE WIDTH OF SOURCE SLIT

A simplification which was made in the above treatment, and which never holds exactly in practice, is the assumption that the source slit (S' of Fig. 16B) is of negligible width. This is necessary in order that the lens shall furnish a single train of plane waves falling on the double slit. Otherwise there will be different sets of waves approaching at slightly different angles, these originating from different points in the source slit. They will produce sets of fringes which are slightly shifted with respect to each other, as illustrated in Fig. 16G(*a*). In the figure the interference maxima are for simplicity drawn with uniform intensity, neglecting the effects of diffraction. Let P and P' be two narrow lines acting as sources. These may be two narrow slits, or, better, two lamp filaments, since we assume no coherence between them. If the positions of the central maxima of the interference patterns produced by these are Q and Q', the fringe displacement QQ' will subtend the same angle α at the double slit as the source slits do. If this angle is a small fraction of the angular separation λ/d of the successive fringes in either pattern, the resultant intensity distribution will still resemble a true $\cos^2 \gamma$ curve, although the intensity will not fall to zero at the minima. The relative positions of the two patterns, and the sum of the two, in this state are illustrated in Fig. 16G, curves (*b*). Curves (*c*) and (*d*) show the effect of increasing the separation PP'. For (*d*) the fringes are completely out of step, and the resultant intensity shows no fluctuations whatever. At a point such as Q the maximum of one pattern then coincides with the next minimum of the other, so that the path difference $P'AQ - PAQ = \lambda/2$. In other words, P' is just a half wavelength farther from A than is P. If the intensity of one set of fringes is given by $4A^2 \cos^2 (\delta/2)$ or $2A^2(1 + \cos \delta)$, that of the other is

$$2A^2[1 + \cos (\delta + \pi)] = 2A^2(1 - \cos \delta)$$

The sum is therefore constant and equal to $4A^2$, so that the fringes entirely disappear. The condition for this disappearance of fringes is $\alpha = \lambda/2d$. If PP' is still further increased, the fringes will reappear, becoming sharp again when α equals the fringe

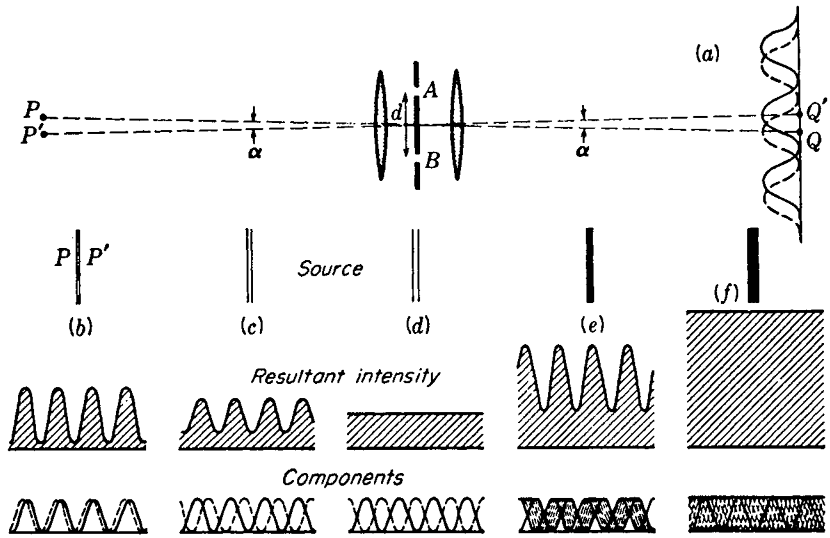

FIGURE 16G
Effect of a double source and of a wide source on the double-slit interference fringes.

distance λ/d, disappear when the fringes are again out of register, etc. In general, the condition for disappearance is

$$\alpha = \frac{\lambda}{2d}, \frac{3\lambda}{2d}, \frac{5\lambda}{2d}, \ldots \qquad \textit{Disappearance of fringes with double source} \qquad (16i)$$

where α is the angle subtended by the two sources at the double slit.

Next let us consider the effect when the source, instead of consisting of two separate sources, consists of a uniformly bright strip of width PP'. Each line element of this strip will produce its own set of interference fringes, and the resultant pattern will be the sum of a large number of these, displaced by infinitesimal amounts with respect to each other. Figure 16G(*e*) illustrates this for $\alpha = \lambda/2d$, that is, for a slit of width such that the extreme points acting alone would give complete disappearance of fringes as in (*d*). The resultant curve now shows strong fluctuations, and the slit must be still further widened to make the intensity uniform. The first complete disappearance will come when the range covered by the component fringes extends over a whole fringe width, instead of one-half, as above. This case is shown in Fig. 16G(*f*), for a slit of width subtending an angle $\alpha = \lambda/d$. Widening the slit still further will cause the fringes to reappear, although they never become perfectly distinct again, with zero intensity between fringes. At $\alpha = 2\lambda/d$ they again disappear completely, and the general condition is

$$\alpha = \frac{\lambda}{d}, \frac{2\lambda}{d}, \frac{3\lambda}{d}, \ldots \qquad \textit{Disappearance of fringes with slit source} \qquad (16j)$$

It is of practical importance, in observing double-slit fringes experimentally, to know how wide the source slit may be made in order to obtain intense fringes without seriously impairing the definition of the fringes. The exact value will depend on our criterion for clear fringes, but a good working rule is to permit a maximum discordance of the fringes of about one-quarter of that for the first disappearance. If f' is the focal length of the first lens, this corresponds to a *maximum permissible width* of the source slit

$$PP' = f'\alpha = \frac{f'\lambda}{4d} \qquad \text{(16k)}$$

16.8 MICHELSON'S STELLAR INTERFEROMETER

As shown in Sec. 15.9, the smallest angular separation that two point sources may have in order to produce images which are recognizable as separate, in the focal plane of a telescope, is $\alpha = \theta_1 = 1.22\lambda/D$. In this equation [Eq. (15k)] D is the diameter of the objective of the telescope. Suppose that the objective is covered by a screen pierced with two parallel slits of separation almost equal to the diameter of the objective. A separation of $d = D/1.22$ would be a convenient value. If the telescope is now pointed at a double star and the slits are turned so as to be perpendicular to the line joining the two stars, interference fringes due to the double slit will in general be observed. However, according to Eq. (16i), if the angular separation of the two stars happens to be $\alpha = \lambda/2d$, the condition for the first disappearance, no fringes will be seen. Those from one star completely mask those from the other. Hence one could infer from the nonappearance of the fringes that the star was double with an angular separation $\lambda/2d$ or some multiple of this. (The multiples could be ruled out by direct observation without the double slit.) But this separation is only half as great as the minimum angle of resolution of the whole objective $1.22\lambda/D$, which in this case equals λ/d. In this connection it is instructive to compare, as in Fig. 16H, the dimensions of the diffraction pattern due to a *rectangular* aperture of width b with the interference pattern due to two narrow slits whose separation d is equal to b. The central maximum is only half as wide in the second case. Hence it is sometimes said that the resolving power of a telescope can be increased twofold by placing a double slit over the objective. This statement needs two important qualifications, however. (1) The stars are not "resolved" in the sense of producing separate images, but their existence is merely inferred from the behavior of the fringes. (2) A partial blurring of the fringes, without complete disappearance, can be observed for separations much less than $\lambda/2d$, showing the existence of two stars, and from this point of view the minimum resolvable separation is considerably smaller than that indicated by the above statement. In practice it is about one-tenth of this.

The actual measurement of the separation of a given close double star is made by having the slit separation d adjustable. The separation is increased until the fringes first disappear; then, by measuring d, the angular separation is obtained as $\alpha = \lambda/2d$. The effective wavelength λ of the starlight must, of course, be also estimated or measured. Separations of double stars are not often determined by this method, since observations on the doppler effect (Sec. 11.10) afford an even more sensitive means

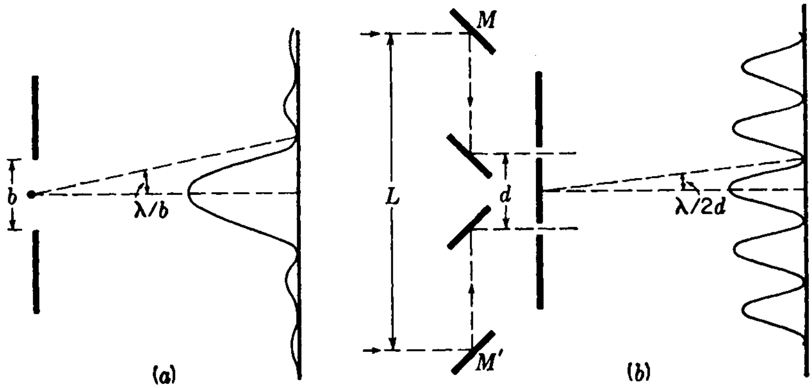

FIGURE 16H
Fraunhofer pattern from (*a*) a rectangular aperture and (*b*) double slit of separation equal to the width of the aperture in (*a*). In (*b*) are shown the four auxiliary mirrors used in the actual stellar interferometer.

of detection and measurement. On the other hand, the method of double-slit interference was, until recently,* the only way of measuring the diameter of the disk of a single star, and in 1920 this method was successfully applied by Michelson for this purpose.

From the discussion of the preceding section it will be seen that if a source such as a star disk subtends a finite angle, disappearance of the fringes would be expected from this cause when the separation of the double slit on a telescope is made great enough. Michelson first demonstrated the practicability of this method by measuring the diameters of Jupiter's moons, which subtend an angle of about 1 second. The values of d for the first disappearance are only a few centimeters in this case, and the measurement could be made by a double slit of variable separation over the objective of a telescope. Because the source is a circular disk instead of a rectangle, a correction must be applied to the equation $\alpha = \lambda/d$ for a slit source. This correction can be found by the same method that is used in finding the resolving power of a circular aperture, and gives the same factor. It is found that $\alpha = 1.22\lambda/d$ gives the first disappearance for a disk source. Estimating the angular diameters of the nearer fixed stars of known distance by assuming they are the same size as the sun, one obtains angles less than 0.01 second. The separations of the double slit required to detect a disk of this size are from 6 to 12 m. Clearly no telescope in existence could be used in the way described above for the measurement of star diameters. Another drawback would be that the fringes would be so fine that it would be difficult to separate them.

Since the blurring of the fringes is the result of variations of the phase difference between the light arriving at the two slits from various points on the source, Michelson realized that it is possible to magnify this phase difference without increasing d. This was done by receiving the light from a star on two plane mirrors M and M'

* See R. Hanbury-Brown and R. Q. Twiss, *Nature*, **178**:1447 (1956).

[Fig. 16H(b)] and reflecting it into the slits by these and two other mirrors. Then a variation of α in the angle of the incoming rays will cause a difference of path to the two slits of $L\alpha$, where L is the distance MM' between the outer mirrors. The fringes will now disappear when this difference equals 1.22λ, and so the sensitivity is magnified in the ratio L/d. In the actual measurements, M and M' were two 15 cm mirrors mounted on a girder in front of the 100-in. Mt. Wilson reflector so that they could be moved apart symmetrically. In the case of the star Arcturus, for example, the first disappearance of fringes occurred at $L = 7.2$ m, indicating an angular diameter $\alpha = 1.22\lambda/L$ of only 0.02 second. From the known distance of Arcturus, one then finds that its actual diameter is 27 times that of the sun.*

16.9 CORRELATION INTERFEROMETER

Another approach to the determination of stellar diameters has been to measure some parameter related to the phase of the incident light. Consider light from a distant source incident at one aperture of the Michelson stellar interferometer. Since the intensity at any time over a given light field is composed of a finite number of random wave trains, or photons, one would expect fluctuations in *phase*, *intensity*, and *polarization*. An abrupt change in intensity would be related to an abrupt change in the makeup of the photon field at the slit, which in turn would likely produce an abrupt change in *net phase*. Similarly a momentary lull in intensity fluctuation would correlate with a nonchanging phase. Therefore one would expect fluctuations in phase to be correlated with fluctuations in intensity. Moreover, the fluctuations would occur at frequencies much lower than the frequency of the light itself.

This correlation of light field intensity with phase, called the *Hanbury-Brown–Twiss effect*, was established by these scientists† through experiment in 1956. The technique ultimately led to a stellar interferometer which far surpasses the Michelson interferometer in resolving distant sources of finite angular size. Its major advantage is that the intensity correlation is not sensitive to slight variations in displacement of the optical components.

The crucial problem at the time of their experiment was to develop a method of measuring the intensity-fluctuation correlation with temporal resolution small enough to detect the fluctuations. The answer to this problem was to use two separated parabolic reflectors focused on photomultiplier tubes (see Fig. 16I). The outputs were delivered to electronic circuitry which produces an output proportional to the product of the two inputs. This in turn was sent through an integrating, or averaging, circuit. The variation of this output with detector separation is called the *second-order interference function* and displays an interference pattern similar to the Michelson interferometer (*first order interference*). With this technique the separation of the detectors can be greatly extended without having the interference pattern destroyed by slight variations in the mirror positions.

* Details of these measurements will be found in A. A. Michelson, "Studies in Optics," chap. 11, University of Chicago Press, Chicago, 1927.

† R. Hanbury-Brown and R. Q. Twiss, Correlation between Photons in Two Coherent Beams of Light, *Nature*, 127:27 (1956).

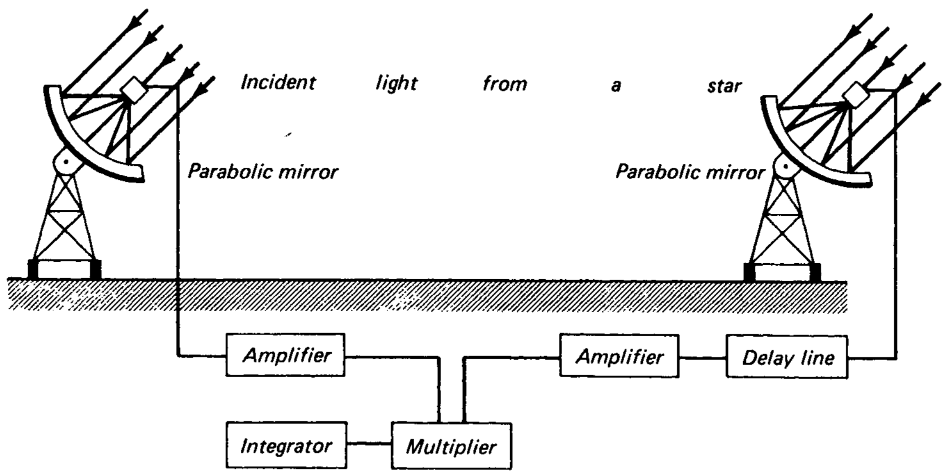

FIGURE 16I
Photoelectric detectors and electronic circuitry for a long-base-line correlation interferometer.

Using ordinary searchlight mirrors to focus starlight onto photomultiplier tubes, Hanbury-Brown and Twiss studied the star Sirius and were able to determine its angular diameter to be 0.0069 second of arc.

Since that time, a correlation interferometer with a base line 188 m long has been built at Narrabri, Australia, where angular diameters as small as 0.0005 second of arc can be measured. This far surpasses the results of the Michelson stellar interferometer.*

16.10 WIDE-ANGLE INTERFERENCE

Nothing has thus far been said about any limit to the angle between the two interfering beams as they leave the source. Consider, for example, the double-slit arrangement shown in Fig. 16J(*a*). The source S could be a narrow slit, but to ensure that there is no coherence between the light leaving various points on it, we shall assume that it is a self-luminous object. It is found experimentally that the angle ϕ may be made fairly large without spoiling the interference fringes, provided the width of the source is made correspondingly small. Just how small it must be is seen from the fact that the path difference from the extreme edges of the source to any given point on the screen such as P must be less than $\lambda/4$. Now if we call s the width of the source,

* For additional reading see W. Martienssen and E. Spiller, Coherence and Fluctuations in Light Beams, *Am. J. Phys.*, **32**: 919 (1964); A. B. Haner and N. R. Isenor, Intensity Correlations from Pseudothermal Light Sources, *Am. J. Phys.*, **38**: 748 (1970); and K. I. Kellermann, Intercontinental Radio Astronomy, *Sci. Am.*, **226**: 72 (1972).

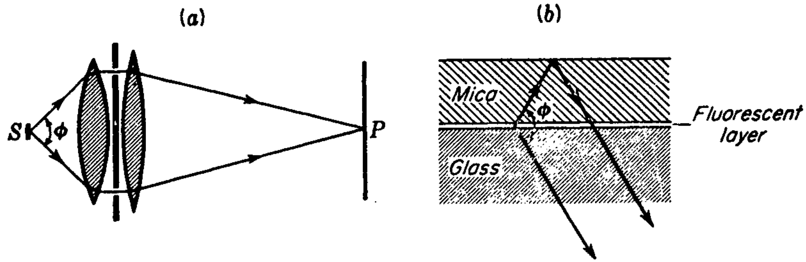

FIGURE 16J
Two methods of investigating wide-angle interference.

the discussion given in Sec. 15.10 shows that this path difference will be $2s \sin(\phi/2)$. Hence, for a divergence of 60°, s cannot exceed one-quarter of a wavelength, or 1.3×10^{-5} cm for green light. If the width is made greater than this, the fringes disappear completely when the path difference is λ, reappear, and then disappear again at 2λ, etc., just as in the stellar interferometer. By using as a source an extremely thin filament, Schrödinger could still detect some interference at an angular divergence ϕ as large as 57°.

An equivalent experiment which permitted using even larger angles of divergence (up to 180°) was performed by Selenyi in 1911. The essential part of his apparatus, shown in Fig. 16J(b), was a film of a fluorescent liquid only one-twentieth of a wavelength thick contained between a thin sheet of mica and a plane glass surface. When the film is strongly illuminated, it becomes a secondary source of light having a somewhat longer wavelength than the incident light (see Sec. 22.6). Interference can then be observed in a given direction between the light that comes directly from the film and that which is reflected from the outer surface of the mica. Interesting conclusions about the characteristics of the radiating atoms, in particular whether they radiate as dipoles, quadrupoles, etc., can be drawn from data on the variation of the visibility of the fringes with angle.*